

# Mahmood Seoud

*E-mail:* mahmoodseoud@gmail.com \* *Telephone number:* +4531603874

## Professional Summary

---

### Embedded Systems Engineering Student

#### Profile

Computer Science Master's student specializing in embedded Linux systems and low-level C/C++ development. Hands-on experience building custom Yocto distributions and real-time communication protocols for resource-constrained environments. Seeking to contribute to Linux-based embedded audio platforms.

## Education

---

### Elite Summer School - Robotics & Entrepreneurship

August 2025

#### Intensive Program

University of Southern Denmark

Collaborative industrial robots, AI algorithms, mobile robot manipulators

### Master's in Computer Science

July 2024 - June 2026

#### 3<sup>rd</sup> semester

IT University of Copenhagen

Focus: Digital fabrication, rapid prototyping, advanced AI/ML applications

### Bachelor's in Computer Science

July 2021 - June 2024

#### Thesis: Mesh Grid Networks

University of Copenhagen

Strong emphasis on algorithms, mathematics, and problem-solving

## Professional Experience

---

### PeopleNet A/S

August 2022 - Present

#### Software Developer (Part-time)

Copenhagen, Denmark

- **Developed enterprise applications** for SharePoint solutions, demonstrating professional software development practices and collaborative teamwork
- **Integrated new features into existing codebases** using Git workflows, maintaining code quality and consistency across multi-developer projects

## Technical Projects

---

### DISCOSAT CubeSat Satellite Platform

January 2025 – Present

#### Embedded Systems Developer (Volunteer)

Copenhagen, Denmark

- **Built custom embedded Linux distribution using Yocto/OpenEmbedded** with stripped kernel and minimal rootfs, reducing boot time by 35% through init system optimization and kernel configuration tuning for ARM architecture
- **Implemented CubeSat Space Protocol (CSP) in C** for real-time telemetry transmission to ground station, including packet routing and error handling in resource-constrained environment

## Technical Skills

---

#### Languages:

C, C++ (C++11/14), Python, Java

#### Embedded Systems:

Yocto/OpenEmbedded, Linux kernel configuration, cross-compilation toolchains, ARM architecture, real-time systems

#### Development Tools:

Git, GCC/G++, Make, CMake, Docker, CI/CD pipelines

#### Protocols & Libraries:

CubeSat Space Protocol (CSP), OpenCV, TCP/IP networking